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Review: Interventions addressing loneliness amongst university students: a systematic review

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Background: Loneliness is detrimental to mental health, with university students at higher risk of feeling lonely than other population groups. However, little research has explored interventions to reduce loneliness among students. This review identifies the characteristics and effectiveness of interventions targeting university/college students. **Methods:** PsycINFO, Medline, ASSIA and Web of Science were searched from inception using keywords linked to 'loneliness', 'intervention' and 'students'. Relevant peer and nonpeer-reviewed English-language articles on studies implementing an intervention with loneliness as an outcome and investigating undergraduate or postgraduate students at a higher education institution were included for quality analysis and narrative synthesis. Risk of bias was assessed at both study level and at outcome level. **Results:** Twenty-eight articles were included, comprising 25 quantitative and three qualitative studies, covering 37 interventions, most implemented in the United States. Interventions were based on psychoeducation, social support groups, increasing social interaction or reflective exercises. The age of the participants (n = 2339) ranged from 17.62 to 25 (mean age 20.63) years. Evidence from the RCTs suggests that most interventions influenced loneliness outcomes, but the magnitude of the benefit is unclear. Across quantitative studies, 80% (16/20) of interventions based on either social support groups, increasing social interaction or reflective exercises, and 50% (7/14) of interventions based on psychoeducation were deemed effective in reducing loneliness. Most interventions measured quantitatively were delivered in a group setting, of which two thirds were considered effective in reducing loneliness scores, regardless of intervention. **Conclusions:** Universities have a choice of interventions to help reduce loneliness among students either on campus or virtually. Ones promoting social connectedness appear to be more successful. More high-quality studies in a larger number of countries are needed, taking vulnerable student groups into consideration.

Key Practitioner Message

- Young people often feel lonely on a regular basis. This is a risk factor for the development of mental health conditions. Mental health services provided by universities and other settings do not meet the demand for support, a factor in declining mental health of young people.
- Interventions that bring students together around an activity, in a support group or to socialise, either in person or virtually, can reduce feelings of loneliness in young people. Meditation or mindfulness may benefit those who prefer not to join groups.
- Universities have the power to tackle loneliness, a major health concern in this population. Practitioners need to be involved in designing, facilitating and maintaining safe spaces and opportunities for students and young people to interact and bond to prevent this significant risk factor evolving into mental health diagnoses in the future.

Keywords: Students; university; college; mental health; loneliness; social connectedness; interventions; prevention

Background

University students have been found to suffer with depression, stress and anxiety (Chen et al., 2013; Dalky & Gharaibeh, 2019; Stallman, 2010; Thorley, 2017). This is being addressed globally by improvements to university mental health services (Warwick, Maxwell, Statham, Aggleton, & Simon, 2008). Despite this, college students in the United States (US) reporting past-year treatment (therapy/counselling and/or psychotropic medication) for mental health concerns increased from 19% to 34% between 2007 and 2017 (Lipson, Lattie, & Eisenberg, 2018). In the United Kingdom (UK), the number of first-year students disclosing a mental health condition has increased almost fivefold (from 0.4% in 2006 to 2% in 2016), and there has been a significant rise in the demand for student support services; in some universities, up to 25% of students are using, or waiting to use, counselling services (Thorley, 2017). Meeting this increasing demand is a challenge (Holm-Hadulla & Koutsoukou-Argyraki, 2015; Hughes, Panjwani, Tulcidas, & Byrom, 2018; Thorley, 2017; Williams et al., 2015). New approaches are needed to improve student wellbeing. A preventative strategy is to focus on student loneliness.

Loneliness is a subjective, negative experience associated with the sense of having fewer social contacts than desired and/or a lack of anticipated level of intimacy in

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relationships (Yanguas, Pinazo-Henandis, & Tarazona-Santabalbina, 2018). Loneliness is associated with greater anxiety, stress and depression in students (Diehl, Jansen, Ishchanova, & Hilger-Kolb, 2018; Mann et al., 2022; Richardson, Elliott, & Roberts, 2017) and has been associated with maladaptive coping styles, sleep problems, lower life satisfaction and suicidal ideation (Kurina et al., 2011; Zhang, Fan, Huang, & Rodriguez, 2018). Loneliness has also been associated with impaired cognitive health (Lim, Eres, & Vasan, 2020). Loneliness impacts brain functionality, with chronic loneliness changing the nature and likelihood of social interactions (Lim et al., 2020).

In 2014–15, 32.4% of students in Germany felt moderately lonely and 3.2% severely lonely (Diehl et al., 2018). In 2018, young people aged 18–22 were identified as the loneliest generation of Americans (Demarinis, 2020). The same year, 9.8% of young people in the UK aged 16–24 years 'often' felt lonely (Snape et al., 2018). A follow-on survey found that over 15% of students reported feeling lonely on a day-to-day basis, with figures worse for Black, Asian and Minority Ethnic (BAME), disabled and international students, and those living at home (WONKHE, 2019). In 2020, students across the globe were affected by the global COVID-19 pandemic. This caused the closure of campuses, the restriction of face-to-face teaching and the cancelling of sports and society events. Students were confined to their accommodation or at home with parents. Opportunities to develop friendships with other students who share interests and hobbies were reduced (Buote et al., 2007). During the first lockdown, young people and students in the UK and U.S. experienced loneliness and suffered from psychological distress more intensely than the general population (McGinty, Presskreischer, Han, & Barry, 2020). Many reported experiencing mental health symptoms (Tang et al., 2022). In the UK, 18–24 year olds were nearly three times more likely than 65–74 year olds to have experienced loneliness 'more often than normal' (Royal Society for Public Health, 2020).

The evidence suggests a greater need for interventions to reduce loneliness in students, both in response to restrictions related to the COVID-19 pandemic but also outside of such restrictions. While charities and health services offer help for those feeling lonely at university, targeted interventions have been developed to attempt to relieve loneliness. Discussion groups allow students to talk with others about the stresses of university, relationships and self-identity (Gold, Bentzley, Franciscus, Forte, & De Golia, 2019). Social skills training enable students to overcome loneliness by learning new skills (Orchard, 1986). Online chat rooms and forums allow students to connect with other students from home (Barber, 2011). Although universities currently offer various relevant interventions that address student loneliness, these have not been evaluated thoroughly or tested among larger population groups. There is a need for a solid research base (Hunt & Eisenberg, 2010).

Masi, Chen, Hawkley, and Cacioppo (2011) grouped loneliness interventions aimed at adults into four categories: improving social skills, enhancing social support, increasing opportunities for social contact and addressing maladaptive social cognition. Maladaptive social cognition can result in, for example, negative evaluations of other people and lack of interpersonal trust. Addressing maladaptive social cognition was found to be the most effective; however, students were not identified as a subgroup. Bessaha et al. (2020) discovered that interventions involving technology and support groups significantly reduced loneliness in various adult groups. Again, students were not identified as a subgroup.

Whilst reviews by Eccles and Qualter (2021) and Osborn, Weatherburn, and French (2021) targeted young people, they focused on a wide age range, 3–25 and 10–25 years, respectively, and a variety of settings. Hickin et al. (2021) reviewed participants between the ages of 8–81 years, and Williams et al. (2021) included studies with no specified age range. The first three of these latest reviews included a small number of studies on undergraduate students (Bruehlman-Senecal et al., 2020; Loucks et al., 2021; Mattanah et al., 2010; Zhang, Fan, et al., 2018), but contained no special focus or recommendations regarding the university setting. Williams et al. (2021) recognised students as a specific population group but, due to concentrating on interventions compatible with COVID-19 shielding and social distancing measures, identified only five studies focusing on students, all interventions involving a psychological or cognitive component (Conoley & Garber, 1985; Dowd et al., 2014; McWhirter & Horan, 1996; Seepersad, 2005; Zhang, Fan, et al., 2018).

University students are in a transitional phase; they often live away from home for the first time in their lives, and have various academic, financial and social pressures and concerns, which sets them apart from many adults (Horgan, McCarthy, & Sweeney, 2013) and other young people (Lim et al., 2020). Whilst students may benefit from interventions identified in the earlier reviews, it is unclear, which are most suitable and effective for students in a university setting. Most previous reviews have contained interventions explicitly targeting loneliness even though there may be broader, 'indirect' approaches to well-being that do not specifically aim to address loneliness but have important impacts on loneliness (Mann et al., 2017). Due to the scarcity of studies targeting students, in this narrative systematic review our aim was to identify, describe and analyse the effectiveness of intervention studies addressing loneliness in university students (either directly or indirectly). The research questions were: (a) What are the characteristics of interventions that address levels of loneliness among students in higher education (university)? (b) What is the evidence for the effectiveness of interventions that alleviate loneliness among university students?

Methods

Protocol and registration

The protocol for this review was registered with the PROSPERO database (CRD42020180867) (Available from: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020180867). This review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement; Moher, Liberati, Tetzlaff, Altman, & Group, 2009).

Search strategy

Searches were conducted on four electronic databases from inception: PsycINFO, Medline, Web of Science and Applied Social Science Index and Abstracts (ASSIA) on 22nd April 2020. Searches were updated on 22 March 2022. The search period

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was not limited as we were not aware of an earlier review focusing on students. A specialist librarian was consulted regarding databases and the search strategy, which was based on prior research and reviews and combined keywords and MESH terms for the concepts of loneliness, students and interventions (see Table S1).

Reference lists of full text articles were screened for additional relevant studies. Duplicates and nonEnglish titles were removed using a reference management software to provide an initial list of articles. Titles and abstracts were screened independently by two reviewers (OE and CD). Discrepancies were discussed between reviewers, and a third opinion (HT) was sought in the event a discrepancy could not be resolved.

Eligibility criteria and study selection

Journal articles and doctoral dissertations were included only if they met the inclusion criteria specified in Table S2. Studies implementing an intervention with loneliness as an outcome and investigating undergraduate or postgraduate students at a higher education institute were included. 'Higher education' refers to stages 6, 7 and 8 of the International Standard Classification of Education structure, and includes bachelor's or equivalent, master or equivalent and doctorate or equivalent. This excludes all primary and secondary education. We did not limit our results by age of participant as to capture an accurate representation of the student population.

Following the review of titles and abstracts, articles were assessed by reviewing the full text. The inter-rater reliability (IRR) at title an abstract screen was 78.14% (143/183). All 183 articles were included in the full text screen to ensure a thorough process. At full text screen, IRR was 94.5% (173/183). Ten articles with discrepancies were discussed among the research team.

Quality assessment

Risk of bias was assessed at both study level and at outcome level. The revised Cochrane risk of bias tool for randomised trials (ROB2; Sterne et al., 2019) was used to assess randomised experimental studies. The Risk of Bias In Nonrandomised Studies of Interventions (ROBINS-I; Sterne et al., 2016) was used to assess nonrandomised experimental studies. Studies deemed to have 'critical' risk of bias were excluded from data synthesis. Two reviewers (CD and OE) assessed all studies independently then collaborated to identify discrepancies. If disagreements remained, a third opinion was sought (HT).

Data extraction and synthesis

Data extracted from each study included: author, journal, year of publication, location, study type, subgroup, intervention, control, measurement tool, number of participants consented (% male), mean age of participants, % drop-outs preintervention, group sizes at pretest (% male), groups sizes at post-test (% male), group sizes at follow up (% male), % drop out from pretest to follow up, mean age (range) at follow up, measurement points, baseline and postintervention (follow-up) outcomes values, effect sizes (calculated difference in mean scores), *p*-values and evidence of a difference. A pilot extraction was performed on a small sample.

Attempts were made to contact five authors in the event of missing data, e.g., on planned analysis, but these were unsuccessful. All data was extracted by two reviewers (OE and CD) independently, then results were compared for discrepancies. A third opinion (HT) was sought in the event of a disagreement. Given the use of two reviewers, blinding was not performed. An Excel spreadsheet was used to extract data; no other data management tools were used.

A meta-analysis was not possible due to the heterogeneity of studies. Instead, a narrative synthesis of findings was performed to investigate the heterogeneity across the primary studies. Our reporting takes into consideration the Synthesis Without Meta-analysis (SwiM) reporting guidelines (Campbell et al., 2020).

Results

Study selection

Following removal of nonEnglish language and duplicate articles, 6917 articles remained for screening by title and abstract (See Flow diagram in Figure S1). 182 articles underwent full-text screening during which 155 were excluded, leaving 27 articles. Four additional articles were included after searching reference lists of full-text articles. Twenty-eight studies measuring loneliness quantitatively and three studies assessing loneliness qualitatively were included in the quality assessment.

Findings from studies using quantitative measurement of loneliness

Study quality. Mattanah, Brooks, Brand, Quimby, and Ayers (2012) was found to be a secondary analysis of a study already included in our review (Mattanah et al., 2010), so was not assessed. Quality assessment of studies using quantitative data collection is presented in Figure S2 (randomised control trials, $n = 15$) and Figure S3 (nonrandomised designs, $n = 12$). The studies by Mohamed and Ghany (2012), Brodkin, Shrier, Alger, Layman, and Buxton (1983) and Yildiz and Duyan (2022) were found to have an overall critical risk of bias arising mostly from missing data, or from poor quality reporting and small sample sizes; these articles were removed from further review.

Of the remaining 25 studies, seven were judged as having overall high or serious risk of bias and 10 were judged as having overall moderate risk or some concerns. Of the studies judged to have a high or serious risk of bias, most were due to a lack of blinding of participants. As all measurements of loneliness were self-reported, knowledge of the intervention may have influenced participants' responses. Therefore, findings of studies judged as having high or serious risk of bias were interpreted cautiously. All studies other than Mascaro et al. (2018), in which all analyses were conducted using blinding, were judged as either moderate risk or some concerns for risk of reporting bias. No further articles were removed due to concerns of risk of bias.

Study characteristics. Of 25 studies using quantitative measurements of loneliness, 16 were randomised control trials (RCT), three were controlled pretest, post-test designs (CPP) and six were pretest, post-test (PP) designs without a control group. Seventeen of the 26 articles were published within the past 10 years, the remainder between 1985 and 2011.

The characteristics of each study are presented in Table S3. The majority of the studies were conducted in the U.S. ($n = 14$), three in China (Lu et al., 2020; Xiao et al., 2021; Zhang, Fan, et al., 2018), two in Canada (Dowd et al., 2014; Orchard, 1986). One additional study was conducted in each of the following countries: Iran (Yousefipour & Samarin, 2021), Ireland (Worobetz et al., 2020), Japan (Kotera & Fido, 2022), UK (Gee, Hawes, & Cox, 2019), Turkey (Kocak, 2008). All participants were university students; however, some articles specified subgroups, including students with autistic spectrum disorder (Hillier et al., 2018), graduate students (Duncan & Weissenburger, 2003), medical students (Gold et al., 2019; Mascaro et al., 2018; Worobetz et al., 2020), elementary education students

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(Kocak, 2008), psychology students (Besse, 2016; Smith, Ford, Erickson, & Guzman, 2020), first year students (Bruehlman-Senecal et al., 2020; Mattanah et al., 2010; Worobetz et al., 2020), students displaying elevated levels of loneliness at baseline (Bruehlman-Senecal et al., 2020; Dowd et al., 2014; Gee et al., 2019; McWhirter & Horan, 1996; Orchard, 1986; Seepersad, 2005; Yousefipour & Samarin, 2021; Zhang, Fan, et al., 2018), inactive students (Dowd et al., 2014), students with problematic smartphone use (Lu et al., 2020; Xiao et al., 2021), or students self-identifying as stressed, anxious or depressed, or with a pre-existing mental health condition (Gee et al., 2019). The mean ages of participants ranged from 17.62 (Mattanah et al., 2010, 2012) to 25 years (Worobetz et al., 2020); the overall mean was 20.63 (n = 2339). Seven studies reported a mean age of either control or test group <20 (Besse, 2016; Bruehlman-Senecal et al., 2020; Lu et al., 2020; Mattanah et al., 2010; Seepersad, 2005; Xiao et al., 2021; Zhang, Fan, et al., 2018).

Outcome measures. Tools used to measure reductions in feelings of loneliness were relatively consistent across the research, with most (n = 12) using the Revised UCLA Loneliness Scale or UCLA3. The UCLA3 consists of 20 items assessing subjective feelings of loneliness and feelings of social isolation, using questions such as "How often do you feel close to people?". Participants rate each item on a scale of 1 (Never) to 4 (Often) (Russell, 1996). Shorter/longer versions of UCLA were used in some studies (Barber, 2011; Bruehlman-Senecal et al., 2020; Dowd et al., 2014; Kocak, 2008; Lu et al., 2020; Worobetz et al., 2020; Xiao et al., 2021). Other tools used to measure feelings of loneliness included one item of the PERMA profiler (Smith et al., 2020), a three-item loneliness Likert scale (Besse, 2016), The University of the Philippines Loneliness Assessment Scale (UPLAS) (Stewart, Dispenza, Parker, Chang, & Cunnien, 2014), the Chinese College Student Loneliness Scale (CCSLS; Zhang, Fan, et al., 2018) and the short version of the Social and Emotional Loneliness Scale for Adults (SELSA-S; Yousefipour & Samarin, 2021). Measurement points varied. A pre- and post-test measurements only were taken in 11 studies. An additional midtest measurement was implemented in seven studies (including Barber (2011) in which multiple midtest measurements were taken). A further follow-up measure was taken in eight studies. The length of time between post-test measurements and follow-up measurements varied from 2 weeks (Conoley & Garber, 1985; Dowd et al., 2014) to around 6 months (Gold et al., 2019; Mattanah et al., 2010; Seepersad, 2005).

Intervention characteristics. Eighteen studies explored the effectiveness of a single intervention whereas seven studies compared the effectiveness of multiple interventions. In total, 25 studies covered 34 interventions (interventions have been labelled numerically in Table S4). While the aim of eight interventions was solely to reduce levels of loneliness (5, 6, 7, 16, 22, 23, 24, 25), most used loneliness as one measure of intervention effectiveness among others, such as happiness (17, 28), anxiety (2, 17, 34), adjustment to college (4), sense of belonging (18), self-esteem (8, 9, 19, 20, 31, 32), confidence (26), overall wellness (1, 27, 30, 33), empathy (30), depression (2, 8, 9, 10, 11, 31), compassion (29), responsibility (21) and smartphone use (12, 13, 14,15).

The nature of the interventions followed common themes: some provided mainly psychoeducation (5, 6, 7, 8, 9, 10, 11, 16, 18, 24, 25, 28) some implemented social support groups in which students could discuss problems or experiences (4, 26, 32, 30), some provided an opportunity for social interaction with other students outside of a support group, with or without a specific activity or topic of discussion (2, 12, 17, 19, 31, 33, 34) and others implemented reflective exercises, such as mindfulness (1, 3, 20, 22, 23, 27) or mindful movements (e.g. yoga or Mind–Body-Qigong exercises) (13, 14, 29). Where an intervention comprised a mix of two or more elements, such as psychoeducation and social support (26), we have classed them according to the main or distinguishing element.

The majority (n = 28) of interventions were implemented in a group setting, whereas a smaller number (n = 6) were implemented on an individual. Two group interventions were delivered online or digitally (19, 31) as were two individual interventions (16, 20). Of the group interventions, group sizes varied from two participants (31) to 112 participants (28).

Time commitments required from students to partake in each intervention varied. One intervention was a three-day retreat (29). Otherwise, the longest duration of any intervention session was 180 min (17), with others lasting between 30 and 120 min. Overall, support group interventions appeared to require longer sessions whereas individual interventions were found to use shorter sessions. One intervention consisted of one session of psychoeducation to the individual, with an expectation that they engage in physical activity over the following 2 weeks. No time commitment was specified (18).

Of the studies that included a control condition (n = 19), most (n = 12) did not implement an intervention for this group. Instead, measurements coincided with those taken from the intervention group. Seven studies, however, created an alternative intervention to act as a control condition. These included the provision of general information (Besse, 2016; Dowd et al., 2014; Mattanah et al., 2010) an educational course not aimed at reducing loneliness (Kocak, 2008; Smith et al., 2020) and a self-help discussion (McWhirter & Horan, 1996). Two studies offered the control group access to the intervention after a period or completion by the intervention group (Bruehlman-Senecal et al., 2020; Yousefipour & Samarin, 2021).

Study findings. Evidence from the RCTs suggests that most interventions had an effect on loneliness outcomes, but the magnitude of the benefit is unclear (see Table S4). These included six group-delivered psychoeducation interventions (5, 6, 8, 9, 15, 21), five reflective exercises (3, 13, 14, 20, 22), one support group intervention (4) and four social interaction interventions (2, 12, 17, 19). Controlled pre–post (CPP) and pre–post (PP) studies also suggest that group-delivered psychoeducation (41), support group (38) and social interaction (31, 33, 34) may be effective in reducing loneliness. Table S4 reports pre and post mean scores and differences in mean scores (post–pre) to aid estimation of direction and size of effect; studies have been grouped based on study

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design and [version of] loneliness measure to aid comparison of scores. The largest reduction in UCLA3 (20 item) loneliness score was observed with group delivered psychoeducation (22) but this may reflect the fact that the study targeted those most lonely. A social interaction intervention involving group-based basketball three times a week resulted in a large reduction in the ULS-8 (UCLA 8 item) loneliness score (12).

Across all study types, 20 out of 28 group interventions reduced levels of loneliness (1, 3, 5, 6, 8, 9, 12, 13, 14, 15, 17, 19, 21, 22, 26, 28, 31, 32, 33, 34). Three out of six individual interventions (performed with or without the use of the internet) reduced loneliness scores, two in the reflection category (27, 20). Of the interventions performed online or digitally, three were effective in reducing loneliness scores regardless of whether they were performed in an online group or individually (19, 20, 31). Two of these interventions involved chatting to another online (19, 31), and the other involved typing journal entries (20).

Seven of 14 psychoeducation interventions were found to be effective (5, 6, 8, 9, 15, 21, 28) as were three of four social support interventions (4, 26, 32), and seven of nine reflective exercise interventions (1, 3, 13, 14, 22, 27, 29). Three reflective exercises utilised meditation and mindfulness (3, 22, 27), two promoted mindful movements (13, 14), one utilised writing in a journal (20) and one promoted a mindful lifestyle (1). Six of seven social interaction interventions reduced loneliness scores (12, 17, 19, 31, 33, 34), one of which utilised non-human interaction with a dog and facilitator (34). Most of the interventions that were not deemed to be effective were in the psychoeducation category, of which, five of eight were studied in 1996 or earlier (7,8, 9, 10, 11). It should be noted that the majority of this evidence came from RCTs (7/12).

Studies involving students with pre-existing mental health problems or feelings of loneliness (diagnosed or self-declared) used a range of interventions including social interaction (2), reflective exercise (22), social support (26) and psychoeducation interventions (5, 6, 7, 8, 9, 10, 11, 16, 18, 21). However, only six interventions targeting these students significantly reduced loneliness score (5, 6, 8, 9, 21, 22).

Although no study directly compared the effect of a single intervention across subgroups, reductions in feelings of loneliness appeared consistent across groups. For example, reductions in loneliness were seen across students of all educational disciplines, in those with autism spectrum disorder (32) and in students from varied geographic locations. Similarly, reductions in feelings of loneliness were reported in studies sampling mostly female students (8, 9) and in studies sampling mostly male students (32).

Findings from studies using qualitative assessment of loneliness

Study quality. Quality assessment of studies assessing loneliness via qualitative methods is presented in Figure S3 ($n = 3$). Two studies (Clark & Brooms, 2018; Goetzel, Shelov, & Croen, 1983) had an overall moderate risk of bias. Horgan et al. (2013) was found to have a serious risk of bias due to missing data. The primary objective was to assess a decrease in depressive symptoms; however, only a small number of participants (7 out of 118) provided forum posts at follow up, some of which related to the reduction of feelings of loneliness. All studies using qualitative methods were judged as having moderate risk of bias due to confounding factors.

Study characteristics. All interventions explored through qualitative methods ($n = 3$) were delivered as part of a group, with one (27) delivered in an online format (see Table S5). One study recruited medical students (Goetzel et al., 1983), one recruited black, male students (Clark & Brooms, 2018) and one recruited students scoring highly on the Centre for Epidemiological Studies Depression Scale (CES-D) (Horgan et al., 2013). Two studies were conducted in the U.S. (Clark & Brooms, 2018; Goetzel et al., 1983) and one in Ireland (Horgan et al., 2013). Mean ages of participants ranged from 20.17 (Clark & Brooms, 2018) years to 24.6 years (Goetzel et al., 1983).

Clark and Brooms (2018) collected data through semistructured, open-ended interviews as opposed to questionnaires or forums (Goetzel et al., 1983; Horgan et al., 2013). Horgan et al. (2013) primarily used quantitative methods to measure outcomes other than loneliness, for example, reductions in depressive symptoms. Reductions in loneliness were commented on qualitatively.

Data were collected at different points across the research. Horgan et al. (2013) adopted pretest and post-test measurements, whereas Clark and Brooms (2018) and Goetzel et al. (1983) collected data post-test only. Although this design prevents a comparison to baseline levels of loneliness, participants were asked to give their opinions of the intervention, not their levels of loneliness.

Intervention characteristics. The types of interventions implemented included support groups and increasing social interaction (Table S5). Support groups included the 'Human Dimension Group' which adopted a face-to-face discussion around student difficulties with support from a facilitator (37). An online forum allowed students to support each other virtually with access to online resources (36). The 'Black Male Scholars Programme' improved social interaction by encouraging students to play sports together and by enlisting speakers on a number of topics (35).

Study findings. Table S6 summarises the findings of interventions using qualitative data collection, with key quotes from participants included in the publications. The results of a study sampling students of black African or Caribbean descent showed that bringing students together who have a shared cultural experience can reduce the feelings of loneliness (35). Feedback for the Black Males Scholar Programme was generally positive and focussed around reducing alienation of ethnic minorities in universities.

Horgan et al. (2013) reported that, although primarily aimed at reducing depressive symptoms, the online forum (36) resulted in participants feeling less lonely due to being able to talk to others in a similar situation to theirs. Being able to tell someone about their problems improved feelings of loneliness: "It's about empathy and the realisation that you're not alone. Being able to tell someone/people, even if it is just anonymously putting it

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out there into the virtual domain, is supposed to be cathartic in some way. And it kinda' is". Goetzel et al. (1983) reported that support groups for medical students (37) were beneficial for fulfilling social affiliation needs, finding friends, feeling less lonely and for developing a feeling of belongingness.

One theme identified across the qualitative findings was that participants felt 'less alone' as a result of the interventions. Interventions brought students together of similar backgrounds, increasing a sense of belonging, whilst provided opportunities to share experiences. This could suggest that targeting perceived isolation reduces loneliness in students.

Discussion

This systematic review has investigated the characteristics and effectiveness of interventions addressing loneliness in university students. Loneliness is an indicator of wellbeing and may lead to the development of mental disorders if left untended (Mushtaq, Shoib, Shah, & Mushtaq, 2014). Thirty-four interventions addressing loneliness among university students were included, as part of 25 studies. Only four studies, measuring a total of eight interventions, targeted loneliness as a primary outcome (Besse, 2016; Bruehlman-Senecal et al., 2020; McWhirter & Horan, 1996; Zhang, Fan, et al., 2018). Others had a broader wellbeing remit, some including a number of components. Evidence from the RCTs suggests that most interventions had a beneficial effect on loneliness outcomes, including ones in the psychoeducation, reflective exercise, social interaction or social support group categories. Two thirds of interventions conducted in a group setting and measured quantitatively reduced loneliness scores. The measure of loneliness used did not appear to affect the success of the intervention, a finding reported also by Eccles and Qualter (2021). Studies collecting data on loneliness via qualitative assessment highlighted that the sharing of experiences with students of a similar background reduced feelings of loneliness. However, none of the studies followed up participants longitudinally. It is, therefore, unclear whether the interventions had lasting impact. As blinding to the intervention was not possible, all included studies scored with at least 'some concerns' or 'moderate risk of bias' during quality assessment.

The current review represents an unexplored avenue in loneliness research: 16 of 25 articles focusing on university students were published within the past 10 years, the remainder being conducted earlier, over a period spanning 27 years. The majority of the research has been carried out in the U.S., including three of the studies with a narrow focus on loneliness, despite increased numbers of students all over the world suffering from loneliness, especially owing to COVID-19 pandemic.

The four major intervention strategies identified (psychoeducation, increased social support, reflective exercises and increased social interaction) closely mirror those identified by Masi et al. (2011) and Osborn et al. (2021), with 'psychoeducation' being broader but encompassing the 'addressing maladaptive social cognition' strategy. Williams et al. (2021) grouped loneliness into six categories based on a framework by Gardiner, Geldenhuys, and Gott (2018), with social facilitation and animal therapy categorised separately. In the current review, these were both considered to increase social interaction.

Masi et al. (2011) found addressing maladaptive social cognition was the most effective technique. Williams et al. (2021) found 'psychological' interventions to be effective for students, such as mindfulness-based therapies, akin to 'reflective exercises' here. However, their included interventions had to comply with COVID-19 social distancing measures. Most interventions included in the review by Eccles and Qualter (2021) reduced loneliness among young people. In RCTs, interventions focusing on learning a new hobby or skill yielded the largest effect, followed by interventions aimed at social skills training. Notably, most young people included in the review were part of at-risk clinical samples or at-risk nonclinical samples (Eccles & Qualter, 2021). In the current review, 80% of interventions evaluated quantitatively, utilising social support, reflective exercises or increasing social interaction, were effective. Psychoeducation type interventions failed to show similar efficacy; only 50% of interventions were deemed to be effective. The majority of the psychoeducation interventions in our review not showing a positive effect (7/14) were tested in RCTs.

In support of psychoeducational techniques, Jones, Hobbs, and Hockenbury (1982) concluded that loneliness results from a lack of teachable social skills. A review of elements of loneliness interventions focusing on behavioural and cognitive responses and/or coping strategies identified that social skills and communication training was the most frequently employed elements used to address putative mechanisms, such as negative interpersonal appraisals, anxiety and social skills deficits (Käll et al., 2020). Cognitive behavioural therapy was used by some studies either as the treatment format or as a theoretical base for the intervention (Käll et al., 2020). Hickin et al. (2021) reported psychological interventions to be beneficial but made no comparison to other types of interventions. Psychoeducation interventions identified in this review included cognitive restructuring exercises and developing (social) skills with which to tackle loneliness, for example, through roleplaying or gamification. Four effective interventions out of 10, targeted at students with elevated loneliness, were based on psychoeducation. Other interventions were not solely focused on loneliness but had a broader wellbeing remit (Conoley & Garber, 1985; Orchard, 1986; Smith et al., 2020). This may account for some of the variance between the current findings and those reported by Masi et al. (2011), whose review included only interventions targeting loneliness. Our findings suggest that psychoeducation alone, without an opportunity to increase social connections, may not be sufficient for students living in new social surroundings. For example, one of the smartphone app users (Bruehlman-Senecal et al., 2020) would have liked the app to support connections with others due to their difficulties with face-to-face introductions.

Social support groups allowed students to compare experiences, discuss worries, manage stressors of university and seek advice and support. Noticeably, an intervention focused on creating a support group to build friendships in the first year of university demonstrated a reduction in loneliness in students with a mean

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age of under 18 years (Mattanah et al., 2010, 2012). Bessaha et al. (2020) found support groups to be one of the most effective interventions for reducing loneliness in adults. Feeling 'listened to' increased self-esteem and perceived support, both of which mediate loneliness (Yıldız & Karadaş, 2017). Feeling part of a community could also reduce loneliness (Lin, Dean, & Ensel, 2013).

Interventions increasing social interaction encouraged students to spend more time with others outside of a support group, for example through playing sports or partaking in creative exercises. It appears that the act of coming together with other students helps reduce loneliness. Nyqvist, Victor, Forsman, and Cattan (2016) showed young people who simply have more contact with others experience reduced levels of loneliness. In support, a sense of social connectedness has been identified as a protective mechanism against developing mental disorders (Hawkley & Cacioppo, 2010). The promotion of friendships between adolescents has been shown to halve the risk of developing depression over 6 to 12 months (Hill, Griffiths, & House, 2015).

Virtual platforms, such as online chat rooms or smartphone apps, allow social connections to develop remotely. Our results showed that online interventions reduced student loneliness (Barber, 2011; Shaw & Gant, 2002), suggesting virtual platforms to be beneficial. However, literature regarding the impact of technology, particularly of social media, on loneliness is mixed (Hunt, Marx, Lipson, & Young, 2018; Pittman, 2015; Yang, 2016). While promoting social connectedness face-to-face seems to reduce student loneliness, perhaps only specific forms of virtual platform, like those identified here, have a positive impact. Nevertheless, virtual interventions showed exciting potential and may require less resources than face-to-face interventions.

Reflective exercise interventions, such as meditation, journal writing or mindful movement, encouraged students to reflect on their own feelings without interference, differentiating them from cognitive restructuring. Negative thought processes associated with loneliness can be addressed by techniques such as increasing awareness of self and surroundings (Cacioppo, Grippo, London, Goossens, & Cacioppo, 2015). Other documented benefits of mindfulness include reduction of anxiety, depression and mood disorders (Howarth, Smith, Perkins-Porras, & Ussher, 2019). Whilst many of the reflective exercise interventions were implemented among groups of students, supporting social connectedness, reflective exercise interventions can be performed alone. Hence, they may be more useful when group meetings are not possible.

In support of group-based interventions, these alleviate the loss of meaningful social group memberships that results from moving to university. As explained by Haslam et al. (2022), group memberships provide people with social identity, which refers to a person's sense of self that derives from their membership in one or more social groups. Social identification supports the alignment with group norms and the collaboration with others, i.e., it underlies meaningful group-based social connections. Social identification with one or more groups boosts self-esteem and personal control and provides meaning and purpose. Social connections made in groups (e.g., whether associated with mindful movement, a more interactive pursuit or a support group) thus provide young people a sense of shared meaning, purpose, support and efficacy (Haslam et al., 2022). Furthermore, social connectedness can reduce depression and anxiety symptoms for those with (emerging) mental health problems, which further protects against loneliness (Saeri, Cruwys, Barlow, Stronge, & Sibley, 2018). The subjective sense of connectedness to a group has also been posited as the 'active ingredient' in social relationships (Saeri et al., 2018), which may explain why group-based interventions reduce loneliness for students who may otherwise lack close friends at university.

Strength and limitations

To our knowledge, this is the first systematic review to explore interventions addressing loneliness in university students. Following the original submission of this systematic review, two systematic reviews of interventions aimed at reducing loneliness in young people have been published (Eccles & Qualter, 2021; Osborn et al., 2021). However, neither draw special attention to university students.

The scope of our search included younger university students, from the age of 17, to graduate university students. As there is some overlap in age with college students (aged 16–18 years in the UK), our findings may also be applicable to those in postsecondary educational settings.

Our search strategy incorporated a range of search terms, multiple databases and was not limited by the date of publication. Two independent reviewers completed both screening and quality assessment of all research ensuring reliability. Whilst previous systematic reviews have focused on interventions with loneliness as the primary outcome, our review included studies with loneliness as either primary or secondary outcome, thus capturing all relevant research in this area and drawing attention to interventions, which may serve multiple purposes.

The reasons why some interventions were not effective may be because they were not developed specifically to reduce loneliness, and because the studies were not powered to detect differences in secondary outcomes. Grey literature searches were not performed due to time constraints. Although some nonpublished doctoral dissertations were identified, the majority were journal publications, retrieved from databases with mostly peer-reviewed articles. Our review may, therefore, suffer from positive publication bias (Song et al., 2009).

We were unable to perform a meta-analysis due to the heterogeneity and quality of studies. A reduction in loneliness was somewhat comparable between studies using identical measurement tools (e.g., 20 item UCLA3) but was not possible between studies using different tools. Further to this, the UCLA3, the UPLAS Loneliness Scale, Chinese college student loneliness scale and 3 item Likert scale for loneliness were found to have coefficient alphas for reliability of .89 to .94 (Russell, 1996), .93 (Tharayil, 2012), .86 (Zhang, Fan, et al., 2018) and .89 (Besse, 2016), respectively, making any objective comparison difficult.

Interventions were classified under broad groupings, between which will be some overlap as most appeared to include components of multiple categories.

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Interventions were, therefore, categorised based on their most prominent theme introducing an element of subjectivity.

Implications for future research

The majority of research identified was deemed at an increased risk of bias due to quasi-experimental study designs. Further studies adopting a randomised controlled trial design are needed, especially among students with elevated loneliness or mental health problems. Further research is also needed to compare the effect of a given intervention across multiple subgroups, and to evaluate the effectiveness of an intervention longitudinally. Due to the lack of research in BAME students and those with autism or disability, further experimental studies should be conducted in these subgroups. Evidence demonstrates these groups are particularly vulnerable to feelings of loneliness (Kennedy & Barker, 2020; Reed, Giles, Gavin, Carter, & Osborne, 2016). Furthermore, research should take the complexity of loneliness into consideration (Lim et al., 2020), explore in-depth the mechanisms of successful interventions and identify barriers that prevent at risk students from participating in interventions.

Research is shifting focus from traditional methods of targeting loneliness, to the inclusion of digital interventions (Shah, Nogueras, Woerden, & Kiparoglou, 2019). Future research might combine the use of technology with other intervention types, such as facilitating support groups via mobile applications. Virtual communication may remove potential barriers associated with the uptake of interventions, such as physical distance, time/space/facilitator availability and student confidence in meeting strangers.

Lastly, one of the intervention studies (McWhirter & Horan, 1996) differentiated between social loneliness, a lack of a supportive network, and intimate or emotional loneliness, a lack of an intimate relationship, in line with a growing body of research (Hyland et al., 2019; van Tilburg, 2021). Although tailoring interventions to subtypes of loneliness was not effective in this student sample, future studies should take into account the different dimensions of loneliness (Cacioppo et al., 2015). For this purpose, using a loneliness measure that differentiates between social and emotional loneliness, such as de Jong Gierveld loneliness scale, would be beneficial (Penning, Liu, & Chou, 2014). The UCLA measure focuses only on the social dimension.

Conclusions – Implications for universities

Loneliness has detrimental effects to mental and cognitive health and is prevalent in the student population. The adolescent brain continues to mature throughout adolescence up to age of 25 (Arain et al., 2013). Providing support to young students struggling with loneliness may hinder the development of other risky behaviours, e.g. alcohol or drug abuse (Stickley, Koyanagi, Koposov, Schwab-Stone, & Ruchkin, 2014) or internet addiction (Zhang et al., 2018).

COVID-19 related restrictions have worsened the situation for students. Although psychoeducation may be less effective for the general student population than for at risk or clinical populations in other settings and/or age groups, interventions that boost social skills and address maladaptive social cognitions are more relevant than ever. Young people entering university now and over the next few years have had fewer chances of interacting with peers and building quality relationships in shared physical spaces. Educational institutions catering for 16–18-year-olds would be well placed to support this process.

University campuses are ideally placed to tackle loneliness among students as a preventative or targeted strategy and a solution to stem the sharp rise in the demand for mental health/support services. Preventing or reducing loneliness may also have a positive impact on performance figures, such as educational attainment, completion rates and employment prospects (Matthews et al., 2019, 2021).

The current review shows that universities can choose from a variety of strategies. Most successful interventions among students foster social connectedness and are conducted in a group setting. Importantly, interventions need not only focus on, or be advertised as focusing on, loneliness but can have a broader wellbeing remit. Measures that reduce academic stress are likely to support the joining and identifying with social groups (McIntyre, Worsley, Corcoran, Harrison Woods, & Bentall, 2018).

Given the strong social pressure for young people to appear connected (Achterbergh et al., 2020), admitting to loneliness may be difficult (Kerr & Stanley, 2021). Fear of stigmatisation could deter uptake of public interventions. Virtual platforms are, therefore, a promising medium for delivering interventions privately and anonymously or where social gatherings are restricted (Lim et al., 2020).

Further intervention studies should be conducted with loneliness as the primary endpoint in a broader range of countries and cultures, covering students with elevated loneliness, of different ethnic backgrounds and those with autism or a disability.

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Author contributions

H.T. was responsible for the conception and design of the work, and critical revision of the article. O.E. and

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C.D. conducted all activities related to the review, including data collection, data analysis and interpretation, drafting the first manuscript and critical revision of the article, under H.T.'s supervision. All authors approved the final version of the article.

Ethical information

This is a systematic review. No ethical approval or written consent were required.

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Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. Search strategy
Table S2. Inclusion and exclusion criteria.
Table S3. Main characteristics of studies measuring loneliness with a quantitative outcome measure.
Table S4. Summary of the effectiveness of interventions using quantitative measures. Studies have been grouped based on study design and (version of) loneliness measure.
Table S5. Main characteristics of the intervention studies with qualitative data collection on loneliness.
Table S6. Summary of the effect of interventions on student loneliness using qualitative data collection.
Figure S1. Flow diagram.
Figure S2. Quality assessment of randomised control trials producing quantitative data using ROB2.
Figure S3. Quality assessment of nonrandomised control trials producing quantitative data using ROBINS-I.

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